GATE-2009

BT: BIOTECHNOLOGY

Q.1 to Q.7 carry one mark each

1. The method used for prediction of three dimensional structure of a protein from known structure(s) of one or more related proteins is

(*a*) Multiple sequence alignment

(*b*) Homology modeling

(*c*) Phylogeny

(*d*) Docking

2. To produce plants that are homozygous for all traits, the best choice is

(*a*) Protoplast culture

(*b*) Cell suspension culture

(*c*) Anther and pollen culture

(*d*) Apical merisem culture

3. Restriction endonucleases from two different organisms that recognize the same DNA sequence for cleavage are called

(*a*) Isoschizomers (*b*) Isozymes

(*c*) Concatamers (*d*) Palindromes

4. Caspases are involved in the process of

(*a*) DNA replication

(*b*) Mutation and recombination

(*c*) Antibody synthesis

(*d*) Aposptosis

5. Baculovirus expression system is used to express heterologous genes in

(*a*) Mammals (*b*) Plants

(*c*) Insects (*d*) Yeasts

6. A culture vessels in which physical, physicochemical and physiological conditions, as well as cell concentration, are kept constant is known as

(*a*) Cell concentrator

(*b*) Biostat

(*c*) Batch bioreactor

(*d*) Incubator

7. Virus resistant transgenic plants can be developed by the expression of

(*a*) Cowpea trypsin inhibitor

(*b*) Crystalline toxin protein

(*c*) Defective movement protein

(*d*) Snowdrop lectin

Q.8 to Q.21 carry two marks each

8. Which of the following are commonly used reporter genes?

P. NPTgene Q. Luciferase gene

R. CFTR gene S. GFPgene

(*a*) Q, S (*b*) R,S

(*c*) P, R (*d*) P, Q

9. Which of the following statements are true about glyphosate tolerant transgenic plants?

P. Transgenic plants detoxify glyphosate.

Q. Transgenic plants produce an altered enzyme that is not affected by glyphosate.

R. Transgenic plants sequester glyphosate in vacuoles.

S. Transgenic plants overcome the inhibition of aromatic amino acid biosynthesis.

(*a*) P, Q (*b*) R,S

(*c*) Q,S (*d*) P, R

10. Match the item in Group I with an appropriate description in Group 2:

Group 1	Group 2
P. UPGMA	1. Protein sequence database
Q. CLUSTAL	2. Phylogenetic analysis
R. SWISS-PROT	3. 3-D structure visualization
S. RasMOl	4. Multiple sequence alignment

(*a*) P-4, Q-1, R-2, S-3

(*b*) P-2, Q-4, R-1, S-3

(*c*) P-2, Q-3, R-1, S-4

(*d*) P-2, Q-1, R-4, S-3

11. Match the properties in Group 1 with the downstream operation in Group 2:

Group 1	Group 2
P. Size	1. Extraction
Q. Density	2. Distillation
R. Volatility	3. Filtration
S. Solubility	4. Sedimentation

(*a*) P-3, Q-4, R-2, S-1

(*b*) P-4, Q-1, R-2, S-3

(*c*) P-4, Q-3, R-1, S-2

(*d*) P-3, Q-2, R-4, S-1

12. Match the items in Group 1 with their functions in Group 2:

Group 1

P. *rol* genes

Q. Opines

R. Virulence genes

S. *Aux* and *cyt* genes

Group 2

1. Food and energy source
2. Tumor formation
3. Hairy root induction
4. T-DNA transfer and integration

(*a*) P-4, Q-3, R -2, S-1

(*b*) P-3, Q-2, R-4, S-1

(*c*) P-1, Q-3, R-4, S-2

(*d*) P-3, Q-1, R-4, S-2

13. Which of the following statements hold true for pluripotent stem cells (PSCs) under in *vitro* conditions?

P. PSCs can be maintained in an undifferentiated state

Q. PSCs exhibit abnormal and unstable karyotypes

R. PSCs can differentiate into a wide variety of cell types

S. PSCs cannot be passaged continuously

(*a*) P, Q (*b*) P, R

(*c*) Q, R (*d*) Q, S

14. Determine the correctness or otherwise of the following Assertion (a) and Reason (r):

Assertion (a): IPTG (Isopropylthiogalactoside) is a gratuitous inducer of lac operon.

Reason (r) : IPTG is an efficient inducer, but not a substrate of *lac* operon

(*a*) Both (a) and (r) are true and (r) is the correct reason for (a)

(*b*) Both (a) and (r) are true but (r) is not the correct reason for (a)

(*c*) (a) is true but (r) is false

(*d*) (a) is false but (r) is true

15. Which of the following statements are true about bioreactors?

P. Continuous bioreactors provide less degree of control and uniform product quality than batch bioreactors.

Q. Batch bioreactors are ideally suited for reaction with substrate inhibition.

R. Choice of a bioreactor is dictated by kinetic considerations.

S. Fed batch bioreactors are also called semibatch bioreactors.

(*a*) P, Q (*b*) Q, S

(*c*) R, S (*d*) P, R

16. Match the items in Group 1 with correct options in Group 2 :

Group 1	Group 2
P. DNA footprinting	1. Protein-protein interaction
Q. Yeast two-hybrid system	2. VNTR
R. DNA fingerprinting	3. DNA binding protein
S. SAGE	4. Transcriptome analysis

(*a*) P-1, Q-2, R-4, S-3

(*b*) P-3, Q-1, R-2, S-4

(*c*) P-3, Q-4, R-1, S-2

(*d*) P-4, Q-2, R-1, S-3

17. Determine the correctness or otherwise of the following Assertion (a) and Reason (r):

Assertion (a): Bacterial growth is called synchronous when majority of the cells are in same stage of the bacterial cell cycle.

Reason (r) : Synchronous culture can be obtained by growing bacteria in an enriched medium.

(*a*) Both (a) and (r) are true and (r) is the correct reason for (a)

(*b*) Both (a) and (r) are true but (r) is not the correct reason for (a)

(*c*) (a) is true but (r) is false

(*d*) (a) is false but (r) is true

18. Match the products in Group 1 with their possible applications in Group 2:

Group 1	Group 2
P. Erythropoietin	1. Blood clot
Q. Anti-fibrin 99	2. Binding and transport of iron
R. Collagenase	3. Anaemia
S. Transferrin	4. Animal cell separation

(*a*) P-3, Q-1, R-4, S-2

(*b*) P-3, Q-4, R-1, S-2

(*c*) P-2, Q-3, R-1, S-4

(*d*) P-2, Q-1, R-4, S-3

19. Match the products in Group 1 with their producer organisms in Group 2:

Group 1	Group 2
P. Ethanol from glucose	1. Aspergillus niger
Q. Probiotics	2. Leuconostoc mesenteroides
R. Citric acid	3. Saccharomyces cerevisiae
S. Sauerkraut	4. Bifidobacterium

(*a*) P-1, Q-3, R-2, S-4

(*b*) P-3, Q-4, R-1, S-2

(*c*) P-3, Q-4, R-2, S-1

(*d*) P-1, Q-4, R-3, S-2

20. A RNA polymerization assay was performed using ^{3}H UTP as the labelled nucleotide with a specific activity of 500 μCi/ μmol (1 μCi = 2.2×10^6 counts per min). After 10 min incubation, the trichloroacetic acid-insoluble radioactivity was found to be 692521 counts per min as determined in a liquid scintillation counter working at 60% efficiency for ^{3}H. The amount of UTP incorporated into the RNA will be

(*a*) 15 nmol (*b*) 105 nmol

(*c*) 150 nmol (*d*) 50 nmol

21. One unit of glucoamylase enzyme activity is defined as the amount of enzyme required to produce (imol of glucose per min in a 4% solution of Lintner starch at pH 4.5 and 60°C. If in a reaction mixture with 1ml of the crude enzyme preparation containing 8 mg protein and 9 ml of 4.44% starch, 0.6 μmc of glucose/ml-min is produced, what will be the specific activity of the crude enzyme preparation?

(*a*) 1 unit/mg protein

(*b*) 1.5 units/mg protein

(*c*) 0.25 units/mg protein

(*d*) 0.75 units/mg protein

ANSWERS

1. (*b*)	**2.** (*c*)	**3.** (*a*)	**4.** (*d*)	**5.** (*c*)	**6.** (*b*)	**7.** (*c*)	**8.** (*a*)	**9.** (*c*)	**10.** (*b*)
11. (*a*)	**12.** (*d*)	**13.** (*b*)	**14.** (*b*)	**15.** (*c*)	**16.** (*b*)	**17.** (*c*)	**18.** (*b*)	**19.** (*b*)	**20.** (*)
21. (*)									

EXPLANATIONS

1. Homology modeling, also known as comparative modeling of protein, refers to constructing an atomic-resolution model of the "target" protein from its amino acid sequence and an experimental three-dimensional structure of a related homologous protein (the "template"). Homology modeling relies on the identification of one or more known protein structures likely to resemble the structure of the query sequence, and on the production of an alignment that maps residues in the query sequence to residues in the template sequence. It has been shown that protein structures are more conserved than protein sequences amongst homologues, but sequences falling below a 20% sequence identity can have very different structure.

2. In anther culture, the anther swells and dehisces along its upper margin, lengthwise. This phenomenon helps to expose the pollen grain. Alternatively, huge amount of pollen grains can be isolated manually and can be cultured aseptically very easily. Therefore, pollen is more suitable material than egg cell for the production of haploid. The development and production of haploid plant in vitro is very important for the study of fundamental and applied aspects of genetics in the higher plants. Production of homozygous diploid by doubling the chromosome number of haploid in vitro makes a pure line in single step and such homozygous pure line is of great importance in plant breeding. Also it is easy to induce cell division in immature pollen cells in some species. By anther and pollen culture, haploids can be produced in large numbers very quickly.

3. Isoschizomers are pairs of restriction enzymes specific to the same recognition sequence. For example, Sph I (CGTAC/G) and Bbu I (CGTAC/G) are isoschizomers of each other. The first enzyme to recognize and cut a given sequence is known as the prototype, all subsequent enzymes that recognize and cut that sequence are isoschizomers. Isoschizomers are isolated from different strains of bacteria and therefore may require different reaction conditions.

4. Caspases, or cysteine-aspartic proteases or cysteine-dependent aspartate-directed proteases are a family of cysteine proteases that play essential roles in apoptosis (programmed cell death), necrosis, and inflammation.

5. Baculovirus expression in insect cells represents a robust method for producing recombinant glycoproteins. Baculovirus-produced proteins are currently under study as therapeutic cancer vaccines with several immunologic advantages over proteins derived from mammalian sources.

6. Biostat keeps all the given parameters constant.

7. Virus encoded movement proteins (MPs) increase the size exclusion limits of plasmodesmata in inoculated and neighboring cells and facilitate the transport of viral genome to adjacent cells. There are sequence similarities among MPs of different virus groups. Therefore production of transgenic plant expressing dominant negative mutant of MP, i.e., Defective MP, results in resistance to one or more viral diseases by interfering with the function shared by the different viral movement proteins.

7. Cowpea Trypsin Inhibitor: The explanation for this is called co-suppression. The plant has ways of knowing that the viral coat protein should not be produced, and it has ways of eventually shutting down the protein's expression. When the virus tries to infect the plant, the production of its essential coat protein is already blocked.

8. Commonly used reporter genes that induce visually identifiable characteristics usually involve fluorescent and luminescent proteins. Examples include the gene that encodes jellyfish green fluorescent protein (GFP), which causes cells that express it to glow green under blue light, the enzyme luciferase, which catalyzes a reaction with luciferin to produce light, and the red fluorescent protein from the gene dsRed.

9. **Statement Q:** When glyphosphate expressed in chloroplasts of transgenic plants, the insensitive enzymes produced by confer tolerance to the herbicide glyphosphate.

   Statement S: Glyphosphate acts by inhibiting enolpyruvyl-shikimate-3-phosphate synthase (EPSPS), an enzyme in the pathway leading to biosynthesis of aromatic amino acids. Since transgenic plants exhibit resistance to glyphosphate, there is no inhibition of the synthesis of aromatic amino acids.

10. • **UPGMA** (Unweighted Pair Group Method with Arithmetic Mean) is a simple agglomerative or hierarchical clustering method. Inbioinformatics, UPGMA is used for the creation of phenetic trees (phenograms). In a phylogenetic context, UPGMA assumes a constant rate of evolution (molecular clock hypothesis).

    • **CLUSTAL** is a general purpose multiple sequence alignment program for DNA or proteins.

    • **SWISS-PROT** is a curated protein sequence database which strives to provide a high level of annotation (such as the description of the function of a protein, its domain structure, post-translational modifications, variants, etc), a minimal level of redundancy and a high level of integration with other databases.

    • **RasMol** is a computer program written for molecular graphics visualization intended and used primarily for the depiction and exploration of biological macromolecule structures

11. All the properties in column A are phenomena unit for downstream process in column B.

12. **P>3:** The rolA, rolB and rolC genes are plant oncogenes that are carried in plasmids of the plant pathogen Agrobacterium rhizogenes. Following agrobacterial infection, these genes are transferred into the plant genome and cause tumor formation and hairy root disease.

    Q>1: Agrobacterium rhizogenes causes hairy root disease in plants. The transformed roots start producing opines which serve as carbon or nitrogen source for the infecting bacteria.

    R>4: Genes in the virulence region are grouped into the operons virABCDEFG, which code for the enzymes responsible for mediating transduction of T-DNA to plant cells.

    S>2: Agrobacterium rhizogenes transfers a piece of oncogenic DNA, i.e. T-DNA. The T-DNA contains onc genes, the expression of which is responsible for the tumorous character of the transformed plant cells. Three onc genes are of prime importance: cyt, aux1, aux2.

13. In cell biology, pluripotency (from the Latin plurimus, meaning very many, and potens, meaning having power) refers to a stem cell that has the potential to differentiate into any of the three germ layers: endoderm (interior stomach lining, gastrointestinal tract, the lungs), mesoderm (muscle, bone, blood, urogenital), or ectoderm (epidermal tissues and nervous system). Pluripotent stem cells can give rise to any fetal or adult cell type. However, alone they cannot develop into a fetal or adult organism because they lack the potential to contribute to extraembryonic tissue, such as the placenta.

14. IPTG (I-8000, I-8050) is an artificial inducer of the lac operon. IPTG binds to the lac repressor and inactivates it, preventing its binding to the lac operator and thus inducing the expression of beta-galactosidase. IPTG cannot be hydrolyzed and broken down by the cell and is therefore called a "gratuitous" inducer.

15. Bioreactor choice should be dependent upon the kinetic properties if the reactor and its stability condition. Fed batch reactors are semi batch reactor as it holds the nutrient media for a specific retention time.

16. • **DNA footprinting** is a method of investigating the sequence specificity of DNA-binding proteins in vitro. This technique can be used to study protein-DNA interactions both outside and within cells.
    - **Yeast two-hybrid** is based on the reconstitution of a functional transcription factor (TF) when two proteins or polypeptides of interest interact.
    - **VNTRs** have become essential to forensic crime investigations, via DNA fingerprinting and the CODIS database. When removed from surrounding DNA by the PCR or RFLP methods, and their size determined by gel electrophoresis or Southern blotting, they produce a pattern of bands unique to each individual. When tested with a group of independent VNTR markers, the likelihood of two unrelated individuals having the same allelic pattern is extremely improbable.
    - **Serial analysis of gene expression** (SAGE) is a technique used by molecular biologists to produce a snapshot of the messenger RNA population in a sample of interest in the form of small tags that correspond to fragments of those transcripts. It has been successfully used to describe the transcriptome of other diseases and in a wide variety of organisms.

17. **Assertion:** A synchronous or synchronized culture is a microbiological culture or a cell culture that contains cells that are all in the same growth stage.

    Reason: Synchronous cultures can be obtained in several ways:
    - External conditions can be changed, so as to arrest growth of all cells in the culture, and then changed again to resume growth.
    - Cell growth can also be arrested using chemical growth inhibitors.
    - Cells in different growth stages have different physical properties. Cells in a culture can thus be physically separated based on their density or size, for instance. This can be achieved using centrifugation (for density) or filtration (for size).
    - In the Helmstetter-Cummings technique, a bacterial culture is filtered through a membrane. Most bacteria pass through, but some remain bound to the membrane. Fresh medium is then applied to the membrane and the bound bacteria start to grow. Newborn bacteria that detach from the membrane are now all at the same stage of growth; they are collected in a flask that now harbors a synchronous culture.

18. • **Erythropoietin (EPO)** is a hormone that prevents anemia (low blood count) by making red blood cells. Anemia causes fatigue and low energy levels. It occurs when there are not enough red blood cells to carry oxygen from the lungs to supply all the body's needs. Erythropoietin has its primary effect on red blood cells progenitors and precursors by promoting their survival through protecting these cells from apoptosis.
    - **Anti-fibrin 99** is involved in animal cell separation.
    - **Collagenase** enzyme helps in wound healing by assisting in blood clot formation.
    - **Transferrins** are iron-binding blood plasma glycoproteins that control the level of free iron in biological fluids.

19. **P>3:** Saccharomyces cerevisiae is used extensively in batch fermentations to convert sugars to ethanol for the production of beverages and biofuels.

    Q>4: Probiotics are live microorganisms thought to be beneficial to the host organism. Lactic acid bacteria (LAB) and bifidobacteria are the most common types of microbes used as probiotics.

    R>1: In production technique of citric acid, cultures of A. niger are fed on a sucrose or glucose containing medium to produce citric acid.

    S>2: Sauerkraut (sour cabbage) is finely shredded cabbage that has been fermented by various lactic acid bacteria, including Leuconostoc, Lactobacillus, Pediococcus etc.

20. Since $60\% \rightarrow 692551$ cpm

    So, $100\% = 1154201.67\%$ cpm

    Number of moles of ^3H UTP

    $$= \frac{1154201.67\,\text{cpm} \times \mu\,\text{mol}}{500 \times 2.2 \times 10^6\,\text{cpm}}$$

    $$= 1050 \times 10^{-12}\,\text{mole}$$

    $$= 1.05\,\text{nmole}$$

    There is some error in options.